

Complex Numbers

ANSWER KEY



Arithmetic with complex numbers

- 1 Let $z_1 = 4 - 2i$ and $z_2 = -3 + i$. Find $z_1 + z_2$ and $z_1 z_2$.

$$z_1 + z_2 = (4 - 3) + (-2 + 1)i = 1 - i. \quad z_1 z_2 = (4)(-3) - (-2)(1) + [(4)(1) + (-2)(-3)]i = -10 + 10i.$$

- 2 Find $(6 + i) - (2 - 5i)$.

$$= 4 + 6i$$

Complex roots of a quadratic

- 3 Solve $x^2 + 6x + 13 = 0$.

$$\Delta = 6^2 - 4(1)(13) = 36 - 52 = -16 < 0. \text{ By Theorem 1.1, } x = \frac{-6 \pm \sqrt{-16}}{2} = \frac{-6 \pm 4i}{2} = -3 \pm 2i.$$

- 4 Solve $2x^2 - 4x + 10 = 0$.

$$\Delta = (-4)^2 - 4(2)(10) = 16 - 80 = -64 < 0. \quad x = \frac{4 \pm \sqrt{-64}}{4} = \frac{4 \pm 8i}{4} = 1 \pm 2i.$$

- 5 Without solving it, explain how you know $x^2 + 2x + 5 = 0$ has complex (not real) roots.

Its discriminant is $\Delta = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0$. A negative discriminant means the quadratic formula takes the square root of a negative number, giving a complex conjugate pair (Theorem 1.1) instead of real roots.

- 6 Verify that $x = -3 + 2i$ is a root of $x^2 + 6x + 13 = 0$ by direct substitution.

$$(-3 + 2i)^2 = 9 - 12i + 4i^2 = 9 - 12i - 4 = 5 - 12i. \text{ And } 6(-3 + 2i) = -18 + 12i. \text{ Adding: } (5 - 12i) + (-18 + 12i) + 13 = 0. \text{ Yes, it checks out.}$$

Modulus

- 7 Find $|z|$ for $z = -3 + 2i$ (one of the roots found above).

$$|z| = \sqrt{(-3)^2 + 2^2} = \sqrt{13}.$$

- 8 Find $|z|$ for $z = 5 - 12i$.

$$|z| = \sqrt{5^2 + 12^2} = \sqrt{169} = 13.$$

- 9 Explain why a conjugate pair $a \pm bi$ always has the same modulus for both roots.

$|a + bi| = \sqrt{a^2 + b^2}$ and $|a - bi| = \sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2}$ — squaring b or $-b$ gives the same result, so the two moduli are identical.